

DECODED EXPLAINED

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Deadly bird virus PaBV-4 identified for first time in India

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Rahul Karmakar

A team of scientists has, for the first time in India, identified and genetically characterised a virus that often causes rapid death in captive psittacine birds, raising concerns about the health of exotic pets and conservation of threatened parrot species.

Psittacine, which means relating to or resembling a parrot, describes the bird family *Psittacidae* and order *Psittaciformes*. It includes budgerigars, cockatiels, cockatoos, lovebirds, macaws, and parakeets.

The team, led by Pankaj Deka and Sangeeta Das of the Assam Veterinary and Fishery University in Guwahati, found and characterised parrot bornavirus 4 (PaBV-4) circulating among captive psittacine birds in India. Their study was published in *Scientific Reports*.

The other members of the team represent institutions in Assam and Gujarat.

They tested 83 psittacine birds from 13 species in aviaries in Assam, Karnataka, and West Bengal in 2020-2024. The birds in-

cluded the African grey parrot, ochre-marked parakeet, yellow-collared macaw, eastern rosella, blue-crowned hanging parrot, rainbow lorikeet, and scarlet-chested parrot.

Three groups

They divided the birds into three groups: those showing clinical signs of proventricular dilatation disease (PDD), clinically healthy cage mates of suspected cases, and birds that had died of suspected PDD. Caused by avian bornavirus, PDD is a largely fatal viral condition affecting

the digestive and nervous systems of birds.

"Amongst the avian bornaviruses discovered, PaBV belonging to species *Orthobornavirus alphapsittaciforme* possess the highest veterinary relevance and are considered to be a major threat to psittacine aviculture," the researchers said.

"Since the discovery of PaBV in psittacine birds suffering from PDD in 2008, PaBV infections have been reported worldwide."

Upon testing cloacal swabs from live birds and

brain and proventriculus tissue in dead ones, the team found 44 of the 83 birds tested were infected. Infection was detected in almost 88% of the dead birds and 19% of the apparently healthy cage mates, underlining how the virus can lurk without showing outward symptoms. The infected birds belonged to nine species whose conservation status ranged from near-threatened to endangered.

Potential risk

The scientists also found that brain tissue and cloacal

swabs yielded higher detection rates in dead and live birds, respectively. Genetic analysis confirmed all positive samples had PaBV-4, a strain previously reported in Canada, Israel, Japan, South Korea, and the U.S.

The team found no patterns linking genetic clusters to host species, geography or timing. Suggesting this may reflect the global trade in psittacine birds, the researchers said their findings highlight risks to conservation based on breeding captive birds.

SYLLABUS MAPPING

- GS-III, Biodiversity & Environment:
- GS-III, Science & Technology:
- GS-II links through international agreements (CITES) and regulatory bodies

Context

- Indian scientists have, for the first time, found a deadly parrot virus called PaBV-4 (**Parrot Bornavirus 4**) in pet parrots kept in cages in India,
- It exposes big gaps in , how India controls the health, breeding, and trade of exotic (foreign) pet birds.
- The article says infected birds ranged from "near-threatened to endangered." This is worrying because it can wipe out rare, threatened parrot species .

What is IUCN Red List.

- **IUCN** = International Union for Conservation of Nature. **Founded in 1948,**
- The Red List sorts every assessed species into **nine categories**
- **It is an International intergovernmental organization.**

SL NO	Category	Meaning
1	Not evaluated	Not yet studied
2	Data deficient	Not enough information to decide
3	Least concern	Common, wide spread, safe
4	Near threatened	Doing ok now , but soon it becomes at risk soon
5	Vulnerable	High risk of extinction
6	Endangered	Very high risk of extinction
7	Critically endangered	Extremely high risk
8	Extinct in the wild	Survives only in captivity or zoo
9	Extinct	Last individual has gone for ever

Zoonotic vs Non-Zoonotic : Does This Virus Infect Humans?

- **Zoonotic disease:** A disease that can jump from animals to humans (examples: rabies, bird flu, Nipha etc.)
- **Non-zoonotic:** A disease that stays within animals and does not infect humans.
- **Avian bornavirus is NOT considered zoonotic**

CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora):

- It is an international agreement between governments
- To ensure that international trade in wild animals and plants does not threaten their survival
- It came into force in 1975; India joined in 1976
- Now It has 184+ member countries
- CITES is **legally binding** on the countries that have joined it.

It works through three Appendices:

1. **Appendix I** - species most threatened with extinction; commercial international trade is banned
2. **Appendix II** - species not yet threatened but could become so if trade is not controlled - trade allowed with permits
3. **Appendix III** - species a particular country asks others to help protect., (India has listed species here Example - Red Sandalwood)

Prelims PYQ [2015]

With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct?

1. IUCN is an organ of the United Nations and CITES is an international agreement between governments.
 2. IUCN runs thousands of field projects around the world to better manage natural environments.
 3. CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.
-
- A. 1 and 2
 - B. 2 and 3
 - C. 1 and 3
 - D. 1, 2 and 3

UPSC Prelims PYQ 2015:

H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?

- (a) AIDS
- (b) Bird flu
- (c) Dengue
- (d) Swine flu.

First hydrogen train : Why the technology is so challenging

PM to flag off India's first hydrogen train: Why the technology is so challenging

Dheeraj Mishra
New Delhi, July 11

PRIME MINISTER Narendra Modi will flag off India's first hydrogen-powered train in Jind, Haryana on July 17. This will mark a major milestone for the Indian Railways in a green technology where few countries have ventured so far.

According to the Railways, this is one of the world's longest (eight passenger cars and two driving power cars) and most powerful (2400 kilowatt) hydrogen trainsets.

The train can carry at least 682 passengers on the 89-km Jind-Sonapat railway section at a maximum operational speed of 75 km/h. It will make two round trips a day, covering 356 km and consuming an estimated 300 kg of hydrogen.

Hydrogen is highly flammable and needs to be transported at very high pressure. How will the Railways do this? And how will the train work? We explain.

HYDROGEN TRAINS ELSEWHERE

French rolling stock giant Alstom presented the technology for the first time at a 2016 exhibition in Berlin. Its two-car trainset, the Coradia iLint, was launched in Germany two years later, becoming the world's first hydrogen-powered passenger train.

Later, countries such as Japan, China and the US also launched their own hydrogen-powered trains.

The technology, however, is still evolving to cater to the mass transportation of passengers and freight. That is why few countries have hydrogen train services, and those that do

use them mostly for relatively short trips.

HOW THESE TRAINS WORK

Normal electrical locomotives require overhead wires to draw the alternating current that powers their movement. Hydrogen trains don't. They run on the simple formula of combining hydrogen with oxygen to produce the tractive electricity.

In the Indian hydrogen train's case, each of the two driving power cars will have four sets of integrated power packs, comprising hydrogen fuel cells and a lithium ferro phosphate battery. The hydrogen fuel cell will draw in the hydrogen stored on board (440 kg) at high pressure and oxygen from the outside air, then combine them to release electrical energy. One power



The technology, however, is evolving. A handful of countries use them mostly for short trips. PHOTO: NORTHERN RAILWAYS

pack will release 300 kW of energy — 115 kW from the fuel cell and 185 kW from the lithium ferro phosphate battery. The four power packs will provide 1200 kW of power.

Since the train has two power cars, the total power gen-

erated doubles to 2400 kW (3200 hp). This is comparable to the horsepower generated by normal 'electric multiple unit' or 'diesel-electric multiple unit' trains that cover similar, short-haul distances.

A senior Northern Railway

official said that the train was essentially created by replacing the diesel engines on old 'diesel electric multiple unit' rakes with a zero-emission hydrogen-electric propulsion system. The official explained the working of the power pack of the hydrogen cell and battery. "The fuel cell provides a constant power output, meaning it does not fluctuate with the train's changing power demand. When the train starts moving, it initially draws the power from the fuel cell, and since the train's power requirement at that stage is lower, the surplus electricity from the fuel cell goes into charging the onboard battery. As the train reaches higher speed and its power demand increases, the battery kicks in

to provide power in addition to the fuel cell," he said.

"Later, when the train's power requirement goes down, especially when the train is nearing the station, the battery power cuts off and the excess energy of the fuel cell starts charging the battery. Thus,

E. after covering a complete circuit, the battery is almost 80% charged at the end of the journey," said the official.

The main part of the system — the fuel cell — has been imported from the Canadian company Ballard, which specialises in its production.

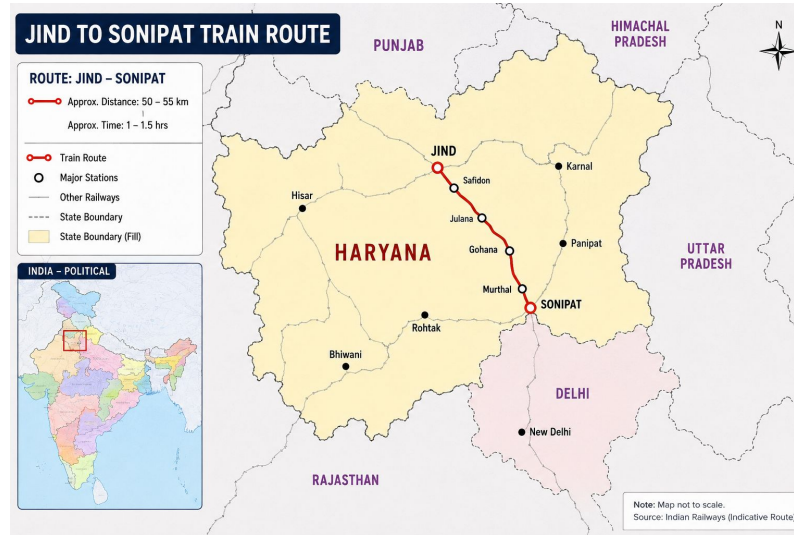
STORING AND SUPPLYING HYDROGEN

Normal atmospheric pressure is measured as one "bar". Hydrogen, however, has to be

stored at 200-500 bar. This is the biggest challenge in using hydrogen as a clean fuel in trains. There are other issues too. Hydrogen production levels are low and transporting is difficult. To ensure the continuous supply of hydrogen for the trains, the railways has set up a 3000-kg-capacity fueling facility at Jind. A chiller plant has also been developed to keep the hydrogen at minus 15 degrees Celsius during dispensing — the low temperature will transform it into a liquid state and, thus, ensure its easy dispensing. The Railways, however, has not yet announced any other hydrogen train project. The official quoted above said that the future of the project would depend on how the first train works. [FULL REPORT ON \[www.indianexpress.com\]\(https://www.indianexpress.com\)](https://www.indianexpress.com)

What's the story ?

- On 17 July 2026, PM Modi will launch India's first hydrogen-powered train, running between Jind and Sonipat in Haryana



Syllabus

Mains (GS Paper III):

Science and Technology — developments and applications

Infrastructure: Energy, Railways

Environment & Ecology - Conservation, environmental pollution and degradation

Every train needs **electricity** to function .There are only **three ways** to get that electricity onto a moving train, and each type of train picks a different way

Electric train / EMU(electric multiple unit): A metal arm on the roof constantly touches overhead wires to draw in electricity, which powers the motors that turn the wheels.

Diesel train/ DEMU(Diesel electric multiple unit): An onboard diesel engine spins a generator to make electricity, which powers the motors that turn the wheels.

Hydrogen train: An onboard fuel cell combines hydrogen with oxygen from the air to make electricity through a clean reaction, which powers the motors that turn the wheels.

How does a hydrogen fuel cell actually work?

- A hydrogen fuel cell is a device that makes electricity through a **chemical reaction** , it takes in **hydrogen** gas (stored on board) and **oxygen** (pulled from the air outside)
- The only "**waste products**" of this reaction are **water**, there's no smoke or carbon dioxide
- That's why it is called **zero-emission technology**

Who's done this before?

- The French train-manufacturing company **Alstom** first showed off this technology at a 2016 **exhibition** in Berlin
- In 2018, its two-car hydrogen train , called the **Coradia iLint** : began running in Germany, becoming the **World's first hydrogen-powered** passenger train.

How the power actually gets used while the train runs

- The **fuel cell always makes the same amount of power**. It never speeds up or slows down what it produces ,it's constant, all the time.
- The battery is different , it can either **take in power or give out power**, depending on what's needed at that moment.

- When the train **just starts moving, it doesn't need much power** yet. But the fuel cell is still making its usual full amount. Since the train isn't using all of it, the **extra power that isn't needed goes into the battery and charges it**.
- As the train speeds up, it needs more power than before. The **fuel cell still only makes its same fixed amount**, which now isn't enough on its own. So the battery **starts releasing the power it stored earlier**, and that power adds on top of the fuel cell's power, so together they give the train enough to go faster.
- As the **train slows down** near a station, it needs **less power again**. The **battery stops releasing** power. Since the fuel cell is still making its full, steady amount, and the train needs less of it now, the **extra power once again goes back into charging the battery**.
- By the end of the trip, all these small top-ups add up, and the battery ends up almost 80% full again ready for the next journey.

Challenges:

- India **doesn't yet produce large amounts of hydrogen**, making a steady daily **fuel supply** genuinely difficult.
- Hydrogen is **highly flammable**
- Hydrogen must be compressed to 200–500 bar (vs. normal air pressure is 1 bar) to fit enough fuel onboard creating major storage and safety challenges. Storing gas at such **extreme pressure demands very strong, specially engineered tanks**, ordinary containers simply can't hold that force safely.
- **The fuel cell** is the train's most critical part, is **imported** from **Ballard**, a **Canadian fuel cell company**.

Different types of hydrogen , based on what type of electricity it uses in electrolysis process

Green: Water split by renewable energy,

Blue: Fossil fuel emissions are captured.

Grey: Natural gas

Brown: Lignite coal is used.

Black: Bituminous coal is used.

Turquoise: Methane is used.

Pink and purple: Water split by nuclear power and heat.

White: Mined directly from underground reservoirs.

Orange: Made from recycled plastic waste.

National Green Hydrogen Mission

- **Targets**: Aims for 5 MMT annual green hydrogen production and 125 GW renewable capacity by 2030.
- **SIGHT Scheme**: Provides financial incentives for domestic electrolyser manufacturing and green hydrogen production.
- **Infrastructure**: Developing dedicated Green Hydrogen Hubs near major industrial clusters and ports for seamless export.

Prelims PYQ 2025

Consider the following types of vehicles:

- I. Full battery electric vehicles.
- II. Hydrogen fuel cell vehicles.
- III. Fuel cell–electric hybrid vehicles.

How many of the above are considered as alternative powertrain vehicles?

- (a) Only one
- (b) Only two
- (c) All the three
- (d) None

UPSC Prelims 2026

Regarding Green Hydrogen, consider the following statements:

1. Produced via natural gas reforming with Carbon Capture and Storage (CCS).
2. Produced via electrolysis using renewable energy.
3. Targets ~50 MMT annual emission reduction by 2030.

Select the correct answer

- (a) 1 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1, 2, and 3

Prelims 2023

With reference to green hydrogen, consider the following statements:

1. It can be used directly as a fuel for internal combustion.
2. It can be blended with natural gas and used as fuel for heat or power generation.
3. It can be used in the hydrogen fuel cell to run vehicles.

How many of the above statements are correct?"

- A) 2 and 3
- B) 1 and 2
- C) None
- D) All three

Navy commissions stealth frigate INS Mahendragiri

Navy commissions stealth frigate *INS Mahendragiri*

V. Kamalakara Rao
VISAKHAPATNAM

The Indian Navy commissioned the sixth advanced P17A-class stealth frigate 'INS Mahendragiri' in Visakhapatnam on Saturday.

The commissioning ceremony, held at the Eastern Naval Command (ENC) headquarters, in the presence of Union Defence Minister Rajnath Singh, marked a significant leap in India's capacity to design, construct, and deploy state-of-the-art naval platforms independently.

Named after the majestic mountain range in the

Eastern Ghats, the vessel symbolises the resilience and indomitable resolve of the Indian armed forces. Its induction into the Navy is viewed as a cornerstone of the 'Atmanirbhar Bharat' initiative, reflecting the country's shift from an importer of military hardware to a global hub for sophisticated warship manufacturing.

An advanced stealth frigate, *INS Mahendragiri* is equipped with cutting-edge technology that places it among the most formidable assets in the Indian Navy's fleet. The vessel features a sophisti-



INS Mahendragiri heading towards Visakhapatnam coast for its official inauguration. SPECIAL ARRANGMENT

cated Combined Diesel or Gas (CODOG) propulsion system, an Integrated Platform Management System

(IPMS), and an array of indigenous combat systems designed for high-intensity naval warfare.

Mr. Singh underscored the balance between embracing next-generation technology and maintaining robust conventional military power. While acknowledging the transformative impact of drones, artificial intelligence, cyber warfare, and hypersonic weapons, he emphasised that the bedrock of national security remains the resolve and expertise of trained soldiers.

"Future wars may be fought with AI, but they will still be won by national resolve, trained soldiers, and credible military power," Mr. Singh remarked.

SYLLABUS MAPPING

GS Paper III - Science and technology

GS Paper II - India and its neighborhood relations

WHY IN THE NEWS

On 11 July 2026, the Indian Navy commissioned INS Mahendragiri, pennant number F38, at the Naval Dockyard, Visakhapatnam, in Andhra Pradesh. It is the sixth of seven Project 17A stealth frigates to be commissioned. The ship joins the Navy Eastern Fleet (the "Sunrise Fleet") based at Visakhapatnam.

Frigate vs destroyer vs corvette vs aircraft carrier

- Corvette: The smallest ocean-going warship , usually for coastal patrolling
- Frigate: Medium and multi-role ship
- Destroyer: Larger and more heavily armed,built for fleet air defence and offensive strike.
- Aircraft carrier: a Floating air base that carries fighter jets.

What is Project 17A (P17A)?

- Project 17A is an Indian Navy programme to build seven advanced stealth frigates called the Nilgiri class
- 17A" means it is the improved "Alpha" version of the earlier Project 17 (Shivalik class).

The seven Nilgiri-class (P17A) ships

1. INS Nilgiri
2. INS Udaygiri
3. INS Himgiri
4. INS Taragiri
5. **INS Dunagiri**
6. **INS Mahendragiri**
7. INS Vindhyagiri

Features on INS Mahendragiri

- The ship can be equipped with the BrahMos surface-to-surface missile
- Indigenous rocket launcher,
- Torpedo launchers
- Integrated Anti-Submarine Defence System

What is "Indigenous content" / "Indigenisation"?

"Indigenous content" means the share of a product (by cost/value) that is made inside India, rather than imported. "Indigenisation" is the process of shifting from importing to making things domestically.

Arguments in favour of Indigenous defence manufacturing

- **Strategic autonomy:** - Availability of spares/weapons during a conflict
- Reduced import dependence and **foreign-exchange savings**
- **Job creation** and MSME involvement
- **Export potential**

Challenges

- **Cost and time overruns:**
- **R&D investment gap:** India's defence R&D spend is small compared to global powers
- **Continued dependence on imported critical sub-systems:** despite the "75% indigenous" label, key items are still foreign — the gas turbines (GE LM2500) and **diesel engines** (MAN 12V28/33D) for propulsion, and important **radar/missile technology** (MF-STAR from Israel, BrahMos co-developed with Russia).

WAY FORWARD

- Deepen indigenisation of critical sub-systems:
- Boost R&D and private participation
- Fix cost/time overruns

What are the changes proposed to US H1B programme?

What are the changes proposed to U.S. H-1B programme?

EXPLAINER

T.C.A. Sharad Raghavan

The story so far: The U.S. government is all set to introduce wide-ranging changes to the rules regarding the manner in which it allows foreigners to work and study in the U.S. – including its H-1B work visa programme, Optional Practical Training (OPT) system, and employment-based Green Cards. These changes are part of recommendations made by three core Departments in the U.S. government – Homeland Security, Labor, and State – under the Unified Regulatory Agendas that they periodically release.

What are the proposed changes to the H-1B system?

The rules are expected to come into effect in August

this year. One such rule, by the Department of Homeland Security (DHS), would reduce the exemptions granted to universities and some research organisations from the cap on H-1B visa issued.

The proposals will also bring in stricter requirements for employers that want to send H-1B workers to third-party client sites. This is of particular importance to Indians since it is a model widely used by Indian IT and consulting firms.

Basically, companies that want to place workers with an H-1B visa in third-party clients will not only need to show that there exists a genuine employer-employee relationship, but also provide evidence that the worker will perform specialised work at the third-party client site. The process will also require more detailed documentation, and companies that



Indians account for 71-74% of approved applications for H-1B visas, capped at 85,000 per year.

have violated the H-1B visa programme's rules in the past would face greater scrutiny.

All of this has major implications for India and Indians. The U.S. has an annual H-1B visa cap of 85,000. Of this, Indians account for about 71-74% of the approved applications.

What changes has the Department of Labor suggested?

According to its proposed changes, the entry-level wage benchmark for the H-1B and Program Electronic Review Management (PERM) system – the first regulatory step for employment-based Green Cards – would be increased from the 17th percentile to the 34th percentile.

Basically, this will hike the higher minimum salary thresholds for employers sponsoring foreign workers, which then would make Green Cards more expensive to pursue.

What changes will students face?

The DHS currently has a visa system for international students based on the 'duration of stay' principle, which means students can stay in the U.S. as long as they keep meeting their

programme's requirements. Now, however, students will be assigned fixed periods of stay, after which they would have to apply for extensions to continue studying or remain in the U.S.

This, again, is a big deal for India and, within that, to two States in particular. According to the latest data, pertaining to 2023-24, India accounted for the largest share of international students in the U.S., having sent about 3.3 lakh students that year. Further, according to the U.S. consular officials, Telangana and Andhra Pradesh accounted for 50% of the students from India.

Apart from this, a separate proposal that is set to come into effect in February 2027 would see tighter rules around the OPT system, including changes to the two-year STEM OPT extension and Curricular Practical Training (CPT)

system. These are two highly popular paths through which Indian students get U.S. work experience after they are done with their degrees.

Are there any other changes on the cards?

A final rule change could increase the paperwork and tension for spouses of H-1B workers who are in the U.S. under the H-4 visa programme. So far, these people could stay and work in the U.S. under the Employment Authorisation Documents (EADs) system, with extensions to this period of stay being granted automatically.

Now, a new change expected this month will see these automatic extensions being stopped. Those holding such visas can of course apply for an extension, but if their renewal is delayed for whatever reason, they could lose their work permit.

H1 B Visas

- The H-1B visa is a **non-immigrant**, employer-sponsored work visa created under the Immigration Act, 1990, that allows U.S. employers to temporarily hire foreign professionals for jobs that require a high level of skill.
- It helps fill skill gaps in sectors such as IT, engineering, finance, and healthcare.
- H1B visas are typically capped each year and allocated through a **lottery system** managed by the U.S. Citizenship and Immigration Services (USCIS).
- Validity: Initially for 3 years, extendable up to 6 years.
- People born in India are the largest beneficiaries accounting for more than 70% of all approved H-1B petitions annually since 2015.

Thank you